Topic 1 — Physics Fundamentals, Measurement and Instruments

Direct Syllabus Mapping: General Science — Physical Quantities, Units, Measurement Systems, Dimensions, Standards, Practical & Astronomical Units, Scientific Measuring Instruments and Apparatus.
Key Must-Score Focus: SI base units (2019 BIPM definitions via fundamental physical constants), 22 coherent derived units with special names, Astronomical Units (Parsec, Light-Year, AU), Pressure conversions (Bar, Pascal, Torr, psi, kgf/cm²), Specialized units (Dobson, Cusec, NTU, Barrel of oil, Horsepower), and Exhaustive Scientific Instruments Directory (Pyrometer, Pyrheliometer, Anemometer, Tachometer, Fathometer, SONAR, RADAR, Polygraph, Accelerometer, Ringelmann Scale, etc.).

Current Affairs

- **BIPM 2019 SI Redefinition (Effective May 20, 2019)**: All 7 base units are now anchored to fixed numerical values of fundamental physical constants:
- Kilogram (kg) is locked to Planck constant h = 6.62607015 \times 10^{-34}\text{ J}\cdot\text{s} (kg\cdot m^2\cdot s^{-1}), terminating the physical Platinum-Iridium prototype cylinder (*Le Grand K*) kept at BIPM Sèvres (near Paris).
- Second (s) is anchored to caesium frequency \Delta \nu_{\text{Cs}} = 9,192,631,770\text{ Hz} (unperturbed ground-state hyperfine transition of Cs-133).
- Metre (m) is anchored to speed of light in vacuum c = 299,792,458\text{ m/s}.
- Ampere (A) is locked to elementary charge e = 1.602176634 \times 10^{-19}\text{ C} (1\text{ A} = 6.2415 \times 10^{18}\text{ electrons/second}).
- Kelvin (K) is locked to Boltzmann constant k = 1.380649 \times 10^{-23}\text{ J/K} (kg\cdot m^2\cdot s^{-2}\cdot K^{-1}), eliminating reliance on the triple point of water (273.16\text{ K}).
- Mole (mol) contains exactly 6.02214076 \times 10^{23} specified elementary entities (Avogadro constant N_A).
- Candela (cd) is locked to luminous efficacy K_{\text{cd}} = 683\text{ lm/W} for monochromatic radiation of frequency 540 \times 10^{12}\text{ Hz}.
- **New SI Prefixes (27th CGPM, Versailles, 2022)**:
- Large: **Ronna** (R = 10^{27}), **Quetta** (Q = 10^{30}). Mass of Earth \approx 6\text{ ronnagrams} (6 \times 10^{24}\text{ kg} = 6 \times 10^{27}\text{ g}).
- Small: **Ronto** (r = 10^{-27}), **Quecto** (q = 10^{-30}). Mass of electron \approx 1\text{ rontogram}.
- **Solar Constant & TSIS-1 (NASA Consensus)**: Total Solar Irradiance (TSI) received at 1\text{ AU} outside Earth's atmosphere is standardized at **1361.6 \pm 0.3\text{ W/m}^2 \approx 1.4\text{ kW/m}^2** (measured by TSIS-1 / SORCE; varies by \sim 0.1\% over the 11-year solar sunspot cycle).
- **MEMS Accelerometers in Modern Tech (IAS 2023 Fact)**: Accelerometers sense dynamic acceleration and tilt. Crucial for: (1) Airbag deployment via instant deceleration detection, (2) Free-

fall sensor (FFS) in laptops parking HDD heads to prevent drive crash, (3) Display auto-rotation between portrait and landscape in smartphones.

1.1 Physical Quantities, Systems of Units & Dimensions

- **Physical Quantity**: Any property of a material or system that can be quantified and measured with an instrument. \text{Physical Quantity} = \text{Numerical Value } (n) \times \text{Unit } (u) Since the magnitude is constant: n_1 u_1 = n_2 u_2. As unit size increases, numerical value decreases proportionally.
- **Fundamental vs Derived Quantities**:
- *Fundamental (Base)*: Independent quantities that cannot be defined in terms of other physical quantities (7 in SI).
- *Derived*: Quantities formed by multiplying or dividing base units with integer/fractional powers.
- **Historical Unit Systems**: 1. *CGS System* (Gaussian metric): Centimetre (length), Gram (mass), Second (time). 2. *FPS System* (British Imperial): Foot (length), Pound (mass; 1\text{ lb} = 453.59237\text{ g}), Second (time). 3. *MKS System* (Giorgi metric): Metre (length), Kilogram (mass), Second (time). 4. *SI System* (1960, 11th CGPM): Comprehensive rationalized MKS system incorporating 7 base and 2 supplementary units.

1.2 Measurement of Length, Mass, Time & Astronomical Distances

- **Julian Year in Astronomy**: Time measurement unit defined as exactly **365.25\text{ days} of 86,400\text{ SI seconds} each** (= 31,557,600\text{ s}). A light-year is the vacuum travel distance of light in one Julian year: \text{Distance} = c \times t = 299,792,458\text{ m/s} \times 31,557,600\text{ s} = 9.460730472 \times 10^{15}\text{ m} \approx 9.461 \times 10^{15}\text{ m}
- **Parsec Geometry**: \text{Angle } \theta = \frac{\text{Arc}}{\text{Radius}} \implies 1\text{ arcsecond} = \frac{1\text{ AU}}{1\text{ Parsec}} Since 1\text{ arcsecond} = \frac{1}{3600}^\circ = \frac{\pi}{180 \times 3600}\text{ radians} = \frac{\pi}{648,000}\text{ rad}: 1\text{ Parsec} = \frac{1\text{ AU}}{\theta} = \frac{648,000}{\pi}\text{ AU} \approx 206,264.8\text{ AU} \approx 3.0857 \times 10^{16}\text{ m} \approx 3.262\text{ light-years}

1.3 Temperature Scales, Thermometry & Pyrometry

- **Liquid Thermometers**:
- *Mercury Thermometer*: Usable range from -39^\circ C (mercury freezing point) to 357^\circ C (boiling point). Can be extended to 500^\circ C by filling nitrogen gas under pressure above mercury column.
- *Alcohol Thermometer*: Usable down to -112^\circ C (ideal for arctic/cryogenic regions where mercury freezes).
- **Total Radiation Pyrometer (High Temperature >1500^\circ C)**:

- Based on **Stefan-Boltzmann Law**: Total radiant energy (E) emitted per unit surface area of a blackbody per second is directly proportional to the fourth power of its absolute thermodynamic temperature: $E = \sigma T^4 \quad (\sigma = 5.670374 \times 10^{-8}\text{ W}\cdot m^{-2}\cdot K^{-4})$
- Since it detects thermal radiation at a distance, it never contacts the source. Used for metallurgical blast furnaces, molten steel, kiln chambers, and stellar bodies.

1.4 Measuring Instruments & Modern Experimental Tools

- **SONAR (Sound Navigation and Ranging)**:
- Emits high-frequency **ultrasonic acoustic waves** into water.
- Velocity of sound in seawater is $\approx 1500\text{ m/s}$ (much faster than 343 m/s in air).
- Distance $d = \frac{v \times t}{2}$ where t is round-trip echo return time.
- *Active Sonar*: Transmits acoustic ping and detects reflected echo.
- *Passive Sonar*: Does not emit sound; only listens to acoustic signatures of other vessels (submarine stealth).
- **RADAR (Radio Detection and Ranging)**:
- Uses pulsed **radio waves / microwaves** ($c = 3 \times 10^8\text{ m/s}$).
- Detects aircraft, missiles, ships, and weather precipitation. Doppler radar additionally measures target radial velocity.
- **Polygraph (Lie Detector)**:
- Invented by John Augustus Larson (1921). Simultaneously tracks 4 autonomic physiological parameters:
  1. Blood pressure (pneumatic cuff), 2. Pulse rate (plethysmograph), 3. Respiration rate (pneumograph tubes), 4. Galvanic Skin Resistance / GSR (sweat gland activation via electrodermal sensors).
- **Ringelmann Smoke Chart**:
- Developed by Maximilien Ringelmann (1888). 5 grids of black lines on white paper representing smoke density from Card 0 (0\% clear) to Card 5 (100\% pitch black). Standard regulatory scale for chimney emission opacity.

1.5 Kinematics Essentials, Average Speed & Impulse Mechanics

- **Average Speed vs Average Velocity**:
- **Equal Distance Segments (Harmonic Mean)**: When a body covers the first half of distance d at speed v_1 and second half at speed v_2, total distance is $2d$ and total time is $t = \frac{d}{v_1} + \frac{d}{v_2} = d\left(\frac{v_1 + v_2}{v_1 v_2}\right)$. $v_{\text{avg}} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{2d}{d\left(\frac{v_1 + v_2}{v_1 v_2}\right)} = \frac{2 v_1 v_2}{v_1 + v_2}$ *(Note: For three equal distances at v_1, v_2, v_3: $v_{\text{avg}} = \frac{3 v_1 v_2 v_3}{v_1 v_2 + v_2 v_3 + v_3 v_1}$).*

- **Equal Time Intervals (Arithmetic Mean)**: When a body travels for time t at speed v_1 and next equal time t at speed v_2: v_{\text{avg}} = \frac{v_1 t + v_2 t}{2t} = \frac{v_1 + v_2}{2}
- **Impulse (J or I) & Linear Momentum (p)**:
- **Definition**: Impulse is the total change in linear momentum produced by a large force acting over a short time interval: \vec{J} = \vec{F} \cdot \Delta t = \Delta \vec{p} = m \vec{v}_f - m \vec{v}_i
- **Units**: Newton-second (N\cdot s) or kilogram metre per second (kg\cdot m/s).
- **Dimensional Formula**: [J] = [F] \cdot [T] = [M^1 L^1 T^{-2}] \cdot [T^1] = [M^1 L^1 T^{-1}] *(Crucial Exam Fact: Impulse and Linear Momentum have the EXACT same dimensional formula [M L T^{-1}] and identical base units).*

1.6 Heat vs Temperature Distinctions, Optics & CGS Conversion Factors

- **Heat (Energy) vs Temperature (Thermal State)**:
- **Heat** is a form of energy transferred across a boundary due to temperature gradient.
  - *Units of Heat*: Joule (J — SI unit), Calorie (1\text{ cal} = 4.184\text{ J}), Kilocalorie (1\text{ kcal} = 1000\text{ cal} = 4184\text{ J}), Erg (1\text{ J} = 10^7\text{ ergs}), British Thermal Unit (1\text{ BTU} \approx 1055\text{ J} \approx 252\text{ cal}).
- **Temperature** is the measure of average kinetic energy of constituent particles.
  - *Units of Temperature*: Kelvin (K — SI base unit), Degree Celsius / Centigrade (^\circ C), Degree Fahrenheit (^\circ F).
  - *Trap Alert*: **Centigrade / Celsius is a unit of temperature, NOT heat!**
- **Power of an Optical Lens**:
- Defined as reciprocal of focal length (f) measured in metres: P = \frac{1}{f\text{ (in metres)}}
- **Unit**: Diopter (D or m^{-1}).
- Sign convention: Convex (converging) lens has positive focal length \implies +D; Concave (diverging) lens has negative focal length \implies -D.
- **Small Length Unit Conversions in Centimetres (cm)**:
- 1\text{ micron } (\mu m) = 10^{-6}\text{ m} = 10^{-4}\text{ cm} = 10^{-3}\text{ mm}.
- 1\text{ nanometre } (nm) = 10^{-9}\text{ m} = 10^{-7}\text{ cm} = 10\text{ \AA}.
- 1\text{ Angstrom } (\text{\AA}) = 10^{-10}\text{ m} = 10^{-8}\text{ cm} = 0.1\text{ nm}.
- 1\text{ Fermi / Femtometre } (fm) = 10^{-15}\text{ m} = 10^{-13}\text{ cm} (Smallest unit of length).
- **Pressure and Force Equivalents**:
- 1\text{ Pascal } (Pa) = 1\text{ N/m}^2.
- 1\text{ bar} = 10^5\text{ Pa} = 10^6\text{ dyn/cm}^2 \approx 1.0197\text{ kgf/cm}^2 \approx 1.0\text{ bar} \approx 1\text{ kgf/cm}^2 (1\text{ kgf/cm}^2 = 98,066.5\text{ Pa}).
- 1\text{ Newton } = 10^5\text{ dynes}.
- 1\text{ Joule } = 10^7\text{ ergs}.

Bilingual Terminology -- Physics Fundamentals and Measurement

English Term	Hindi Term	Key Anchor
Physical Quantity (भौतिक राशि)	भौतिक राशि	Measurable property; scalar (magnitude only) or vector (magnitude + direction)
SI Unit (अंतर्राष्ट्रीय मात्रक प्रणाली)	अंतर्राष्ट्रीय मात्रक	International System of Units; 7 base units
Scalar (अदिश)	अदिश राशि	Has magnitude only; e.g., mass, temperature, speed, energy
Vector (सदिश)	सदिश राशि	Has magnitude + direction; e.g., velocity, force, acceleration
Length (लम्बाई)	लम्बाई	SI unit = metre (m)
Mass (द्रव्यमान)	द्रव्यमान	SI unit = kilogram (kg); measure of matter
Time (समय)	समय	SI unit = second (s); Cs-133 atomic clock
Electric Current (विद्युत धारा)	विद्युत धारा	SI unit = ampere (A)
Temperature (ताप)	ताप	SI unit = kelvin (K); 0 K = absolute zero = -273.15 degC
Luminous Intensity (ज्योति तीव्रता)	ज्योति तीव्रता	SI unit = candela (cd)
Amount of Substance (पदार्थ की मात्रा)	पदार्थ की मात्रा	SI unit = mole (mol); Avogadro's number = 6.022 x 10^23
Parsec (पारसेक)	पारसेक	Largest astronomical distance unit; 1 parsec = 3.26 light-years
Light-Year (प्रकाश वर्ष)	प्रकाश वर्ष	Distance light travels in 1 year; unit of distance NOT time
Astronomical Unit (AU) (खगोलीय इकाई)	खगोलीय इकाई	Mean Earth-Sun distance = 1.496 x 10^11 m
Dobson Unit (डोबसन इकाई)	डोबसन इकाई	Measures ozone layer thickness; normal = 300 DU; hole < 220 DU
Hertz (हर्ट्ज)	हर्ट्ज	SI unit of frequency; 1 Hz = 1 cycle/second
Pascal (पास्कल)	पास्कल	SI unit of pressure; 1 Pa = 1 N/m2
Decibel (डेसिबल)	डेसिबल	Logarithmic unit of sound intensity; 0 dB = threshold of hearing

One-Liner Revision -- Physics Fundamentals and Measurement

#	Fast Fact
1	SI base units = **7**: m, kg, s, A, K, mol, cd
2	Supplementary units = Radian (plane angle) + Steradian (solid angle)
3	New SI prefixes (2022) = Ronna (10^27), Quetta (10^30), Ronto (10^-27), Quecto (10^-30)

#	Fast Fact
4	Light-Year = unit of **distance** (NOT time); 1 ly = 9.46 x 10^15 m
5	1 Parsec = 3.262 light-years = 206,265 AU (coined: Herbert Hall Turner, 1913)
6	1 AU = 1.496 x 10^11 m = ~150 million km (average Earth-Sun distance)
7	Smallest unit of length = **Fermi** (fm = 10^-15 m); used for nuclear radii
8	Angstrom = 10^-10 m; used for atomic radii and wavelengths
9	1 Barrel of oil = **159 litres** (~42 US gallons)
10	1 Nautical mile = **1.852 km**; Knot = 1 nautical mile/hour
11	1 Cusec = 1 ft3/s = 28.317 litres/s (river/canal flow rate)
12	Normal atmospheric pressure = **1 atm = 101,325 Pa = 760 mmHg = 760 torr**
13	1 bar = 10^5 Pa; 1 millibar = 100 Pa (used in meteorology)
14	1 Horsepower = **746 Watts**; Metric HP = 735.5 W
15	1 kWh = **3.6 x 10^6 Joules** (BOT unit; on electricity bill)
16	Dobson Unit: Normal ozone = 300 DU; Ozone hole < **220 DU**
17	Pyrometer = measures very high temperatures (>1500 degC) without contact
18	Anemometer = measures **wind speed**; Wind Vane = measures wind **direction**
19	Barometer: sudden drop = approaching storm; gradual fall = rain; rise = fair weather
20	Hygrometer = measures **humidity**; Hydrometer = measures **specific gravity of liquids**
21	Lactometer = checks **milk purity** (density); Butyrometer = measures **fat in milk**
22	SONAR = uses **ultrasonic waves** (>20 kHz); for submarine/ocean depth detection
23	RADAR = uses **radio/microwaves**; detects aircraft, ships, weather
24	Fathometer = measures ocean depth; 1 fathom = 6 feet = 1.83 m
25	ECG = heart; EEG = brain; EMG = muscles; Polygraph = lie detector
26	1 kg defined by Planck constant h = 6.626 x 10^-34 J.s (since 2019 BIPM)
27	Cs-133 atomic clock = defines the **second** (9,192,631,770 transitions)
28	Kelvin 0 = absolute zero = -273.15 degC (theoretical minimum temperature)
29	Speed of light (c) = 299,792,458 m/s (defines the metre)
30	Mole = 6.02214076 x 10^23 entities (Avogadro's number; defines the mole)